and chooses between delivering the response directly to the user or using specific tools to perform additional actions, based on the LLM's output. If the output involves data retrieval, the agent makes API calls, processes data and converts it into coherent responses.

Large language models (LLMs): LLMs are a type of AI system trained on a large amount of text data that can understand natural language and generate human-like responses. The processed input is fed into the LLM, which generates a response based on its training.

Tool integration: This helps connect the agent with external applications, databases and automation tools to extend its functionality. Key aspects of tool integration include:

- *API integration:* Agents communicate with other software systems.
- *Third party integration:* Agents integrate with NLP systems and ML models to enhance its capabilities.
- *Automation tools integration:* AI agents can automate repetitive tasks to increase efficiency.

Memory: AI agents remember previous interactions, user preferences, and ongoing tasks to provide a personalised and effective user experience. Key memory management characteristics include scalability, privacy, consistency and adaptability.

AI orchestration: AI orchestration involves managing the coordination and interaction between multiple AI agents to achieve specific goals or tasks. The key components of AI orchestration are:

- *Adaptive task management:* System dynamically assigns and reassigns tasks to AI agents based on their capabilities, availability, and current workload.
- *Multi-agent collaboration:* AI agents work together, sharing information and resources to achieve common goals.
- *Performance monitoring:* System continuously monitors the performance of AI agents to ensure they are meeting the desired outcomes and standards.

Data repository: This includes comprehensive data covering both structured and unstructured data sources. It facilitates efficient data management by utilising both centralised and distributed repositories, employing vector stores for quick information retrieval, and leveraging knowledge graphs for contextual reasoning. The data is categorised and organised so that it can be used by an AI agent and models.

Output: This consists of AI insights that are transformed into personalised, context-aware results. These results are continuously updated. The system's knowledge base is also refreshed in the process.

Governance: The AI architecture integrates essential governance and safeguards to ensure safety, compliance, and ethical AI deployment. It ensures AI agents operate safely, securely, and within regulatory boundaries.

Real world use case: Automated claims processing with AI agents

Use cases of AI agents are endless and evolving continuously, and businesses are experimenting with different ways to incorporate them. Here's a scenario where an AI agent helps streamline claims processing operations (Figure 3).

Claim submission: A policyholder visits their healthcare provider for a medical service. After the service, the healthcare provider submits a claim electronically to the enterprise covering details like patient information, treatment provided, and costs incurred.

Initial claim review: An AI agent receives the electronic claim and begins the initial review.

Data verification: The agent cross-references the submitted claim data with the policyholder's insurance plan details and medical history. It verifies the validity of the claim by checking for any inconsistencies or discrepancies, such as duplicate claims or services not covered by the policy.

Fraud detection: Using advanced machine learning algorithms, the AI agent analyses the claim for any signs of potential fraud.

Policy compliance check: The AI agent ensures that the claim complies with the policyholder's insurance plan terms and conditions.

Approval and reimbursement: If the claim passes all the checks, the AI agent automatically approves it. It calculates the reimbursement amount based on the policy terms and initiates the payment process.

Escalation for manual review: If the AI agent identifies any issues or uncertainties during the review process, it escalates the claim to a human claims adjuster for further investigation.

Continuous learning: AI agents use feedback from the manual review process to continuously improve their algorithms and decision-making capabilities.

In summary, AI agents enhance the efficiency, accuracy, and overall effectiveness of the claims processing workflow in a healthcare insurance setting.

Open source platforms and frameworks for developing AI agents

The leading open source frameworks that provide the necessary tools and mechanisms to build, manage, and orchestrate AI agents are:

LangGraph: This framework is designed for building and managing multi-agent systems. It provides tools for defining agents, managing state, and orchestrating complex workflows.

CrewAI: This open source collaborative tool/framework is used for web searching, data analysis, and content generation. It supports dynamic task allocation and collaboration among multiple agents. Agents can review and improve each other's output.